

Output Settings for Publishing

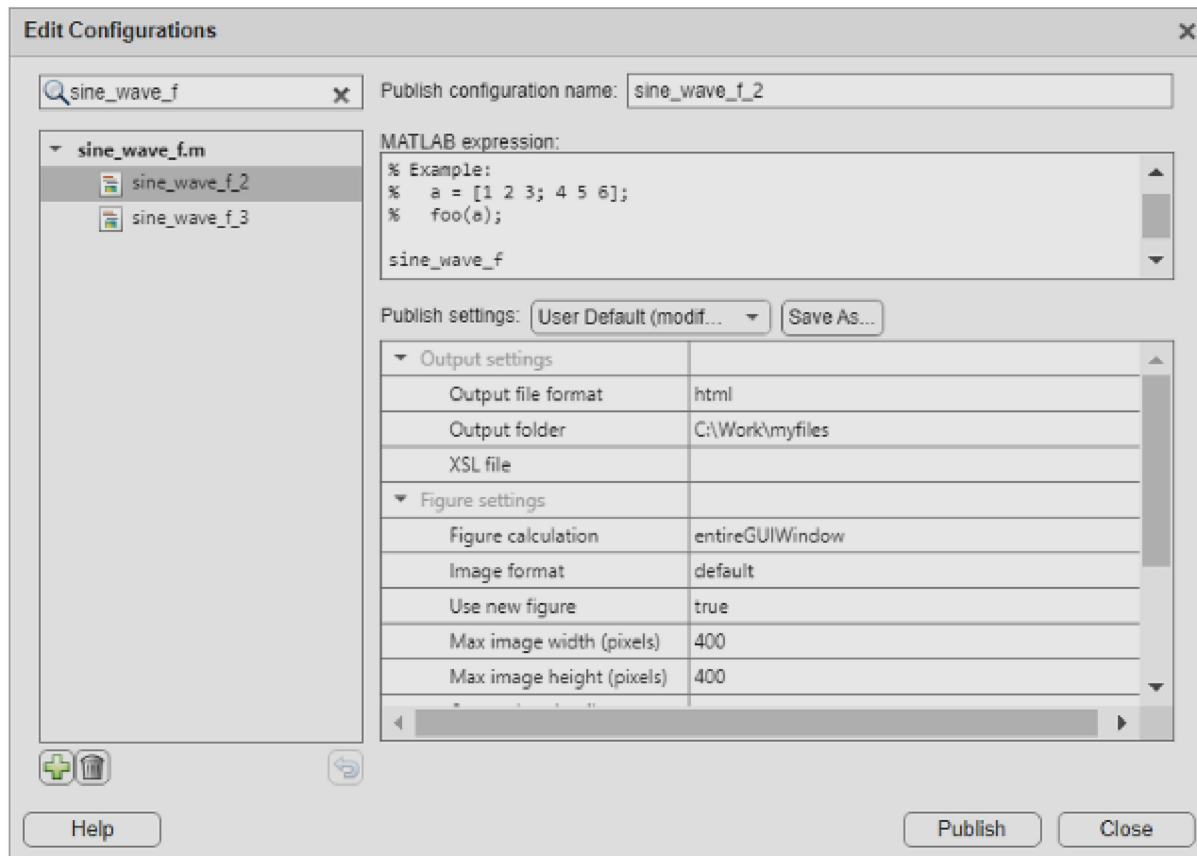
How to Edit Publishing Options

Use the default publishing settings if your code requires no input arguments and you want to publish to HTML. However, if your code requires input arguments, or if you want to specify output settings, code execution, or figure formats, then specify a custom configuration.

1. Go to the **Publish** tab and select **Publish ▼**.
2. Select **Edit Publishing Options**.
3. In the Edit Configurations dialog box, specify output settings.

Use the **MATLAB expression** pane to specify the code that executes during publishing. Use the **Publish settings** pane to specify output, figure, and code execution options.

Together, the panes make what MATLAB® refers to as a *publish configuration*. MATLAB associates each publish configuration with an `.m` file. The name of the publish configuration is displayed at the top of the dialog box and is editable.



Specify Output File

You specify the output format and location on the **Publish settings** pane.

MATLAB publishes to these formats.

Format	Notes
html	Publishes to an HTML document. You can use an Extensible Stylesheet Language (XSL) file.
xml	Publishes to XML document. You can use an Extensible Stylesheet Language (XSL) file.
latex	Publishes to LaTeX document. Does <i>not</i> preserve syntax highlighting. You can use an Extensible Stylesheet Language (XSL) file.
doc	Publishes to a Microsoft® Word document. Does <i>not</i> preserve syntax highlighting. This format is only available on Windows® platforms.
ppt	Publishes to a Microsoft PowerPoint® document. Does <i>not</i> preserve syntax highlighting. This format is only available on Windows platforms.
pdf	Publishes to a PDF document.

Note

XSL files allow you more control over the appearance of the output document. For more details, see <https://docbook.sourceforge.net/release/xsl/current/doc/>.

Run Code During Publishing

- [Specifying Code](#)
- [Evaluating Code](#)
- [Including Code](#)
- [Catching Errors](#)
- [Limiting the Amount of Output](#)

Specifying Code

By default, MATLAB executes the `.m` file that you are publishing. However, you can specify any valid MATLAB code in the **MATLAB expression** pane. For example, if you want to publish a function that requires input, then run the command `function(input)`. Additional code, whose output you want to publish, appears after the functions call. If you clear the **MATLAB expression** area, then MATLAB publishes the file without evaluating any code.

Note

Publish configurations use the base MATLAB workspace. Therefore, a variable in the **MATLAB expression** pane overwrites the value for an existing variable in the base workspace.

Evaluating Code

Another way to affect what MATLAB executes during publishing is to set the **Evaluate code** option in the **Publish setting** pane. This option indicates whether MATLAB evaluates the code in the `.m` file that is publishing. If set to true, MATLAB executes the code and includes the results in the output document.

Because MATLAB does not evaluate the code nor include code results when you set the **Evaluate code** option to false, there can be invalid code in the file. Therefore, consider first running the file with this option set to true.

For example, suppose that you include comment text, `Label the plot`, in a file, but forget to preface it with the comment character. If you publish the document to HTML, and set the **Evaluate code** option to true, the output includes an error.

Modify Plot Properties

```
Label the plot
title('Sine Wave', 'FontWeight','bold')
xlabel('x')
ylabel('sin(x)')
set(gca, 'Color', 'w')
set(gcf, 'MenuBar', 'none')
```

```
??? Undefined function or method 'Label' for input arguments of type 'char'.
```

```
Error in ==> sine_wave2 at 12
Label the plot
```

Use the false option to publish the file that contains the publish function. Otherwise, MATLAB attempts to publish the file recursively.

Including Code

You can specify whether to display MATLAB code in the final output. If you set the **Include code** option to true, then MATLAB includes the code in the published output document. If set to false, MATLAB excludes the code from all output file formats, except HTML.

If the output file format is HTML, MATLAB inserts the code as an HTML comment that is not visible in the web browser. If you want to extract the code from the output HTML file, use the MATLAB [grabcode](#) function.

For example, suppose that you publish `H:/my_matlabfiles/my_mfiles/sine_wave.m` to HTML using a publish configuration with the **Include code** option set to false. If you share the output with colleagues, they can view it in a web browser. To see the MATLAB code that generated the output, they can issue the following command from the folder containing `sine_wave.html`:

```
grabcode('sine_wave.html')
```

MATLAB opens the file that created `sine_wave.html` in the Editor.

Catching Errors

You can catch and publish any errors that occur during publishing. Setting the **Catch error** option to true includes any error messages in the output document. If you set **Catch error** to false, MATLAB terminates the publish operation if an error occurs during code evaluation. However, this option has no effect if you set the **Evaluate code** property to false.

Limiting the Amount of Output

You can limit the number of lines of code output that is included in the output document by specifying the **Max # of output lines** option in the **Publish settings** pane. Setting this option is useful if a smaller, representative sample of the code output suffices.

For example, the following loop generates 100 lines in a published output unless **Max # of output lines** is set to a lower value.

```
for n = 1:100
    disp(x)
end;
```

Manipulate Graphics in Publishing Output

- [Choosing an Image Format](#)
- [Setting an Image Size](#)
- [Capturing Figures](#)
- [Specifying a Custom Figure Window](#)
- [Creating a Thumbnail](#)

Choosing an Image Format

When publishing, you can choose the image format that MATLAB uses to store any graphics generated during code execution. The available image formats in the drop-down list depend on the setting of the **Figure capture method** option. For greatest compatibility, select the default as specified in this table.

Output File Format	Default Image Format	Types of Images You Can Include
doc	png	Any format that your installed version of Microsoft Office supports.
html	png	All formats publish successfully. Ensure that the tools you use to view and process the output files can display the output format you specify.
latex	png or epsc2	All formats publish successfully. Ensure that the tools you use to view and process the output files can display the output format you specify.
pdf	bmp	bmp and jpg.
ppt	png	Any format that your installed version of Microsoft Office supports.
xml	png	All formats publish successfully. Ensure that the tools you use to view and process the output files can display the output format you specify.

Setting an Image Size

You set the size of MATLAB generated images in the **Publish settings** pane on the Edit Configurations dialog window. You specify the image size in pixels to restrict the width and height of images in the output. The pixel values act as a maximum size value because MATLAB maintains an image’s aspect ratio. MATLAB ignores the size setting for the following cases:

- When working with external graphics as described in [External Graphics](#)
- When using vector formats, such as .eps
- When publishing to .pdf

Capturing Figures

You can capture different aspects of the Figure window by setting the **Figure capture method** option. This option determines the window decorations (title bar, toolbar, menu bar, and window border) and plot backgrounds for the Figure window.

This table summarizes the effects of the various Figure capture methods.

Use This Figure Capture Method	To Get Figure Captures with These Appearance Details	
	Window Decorations	Plot Backgrounds
entireGUIWindow	Included for dialog boxes; Excluded for figures	Set to white for figures; matches the screen for dialog boxes
print	Excluded for dialog boxes and figures	Set to white
getframe	Excluded for dialog boxes and figures	Matches the screen plot background
entireFigureWindow	Included for dialog boxes and figures	Matches the screen plot background

i Note

Typically, MATLAB figures have the `HandleVisibility` property set to on. Dialog boxes are figures with the `HandleVisibility` property set to off or callback. If your results are different from the results listed in the preceding table, the `HandleVisibility` property of your figures or dialog boxes might be atypical. For more information, see [HandleVisibility](#).

Specifying a Custom Figure Window

MATLAB allows you to specify custom appearance for figures it creates. If the **Use new figure** option in the **Publish settings** pane is set to true, then in the published output, MATLAB uses a Figure window at the default size and with a white background. If the **Use new figure** option is set to false, then MATLAB uses the properties from an open Figure window to determine the appearance of code-generated figures. This setting does not apply to figures included using the syntax in [External Graphics](#).

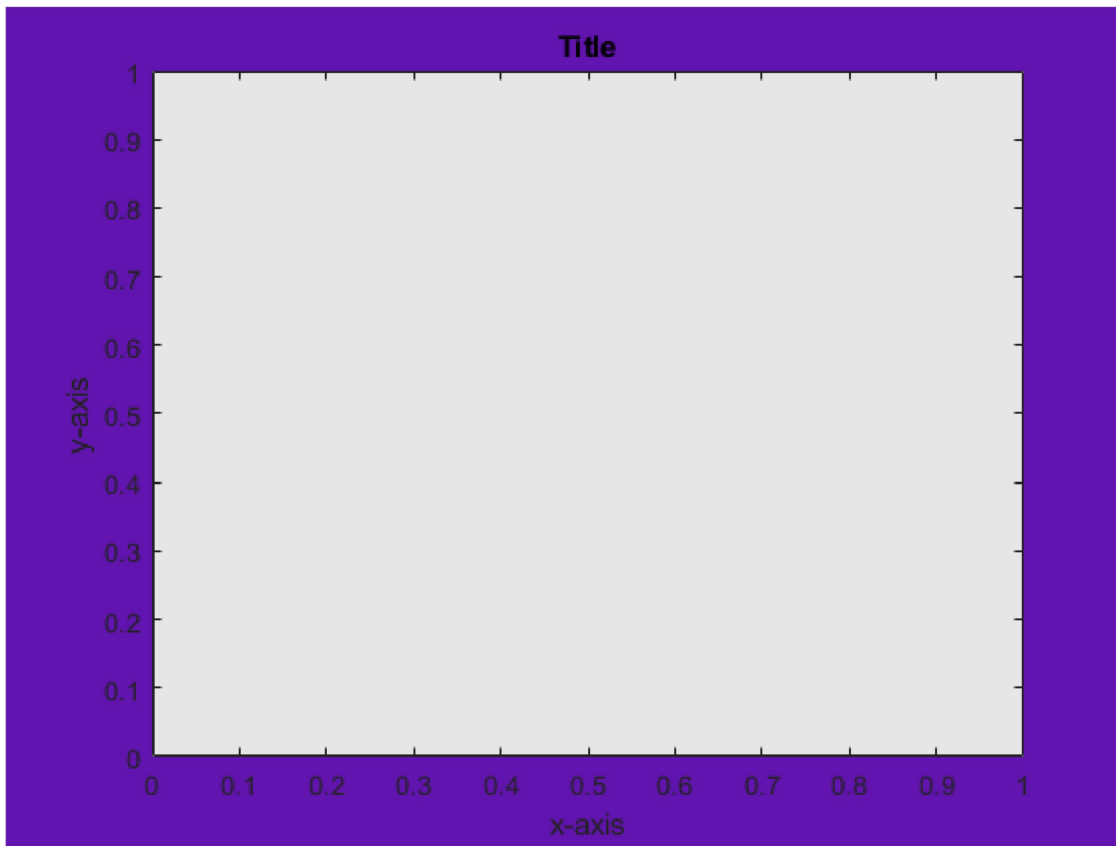
Use the following code as a template to produce Figure windows that meet your needs.

```
% Create figure
figure1 = figure('Name','purple_background',...
'Color',[0.4784 0.06275 0.8941]);
colormap('hsv');

% Create subplot
subplot(1,1,1,'Parent',figure1);
box('on');

% Create axis labels
xlabel('x-axis');
ylabel({'y-axis'})

% Create title
title({'Title'});
```



```
% Enable printed output to match colors on screen  
set(gcf, 'InvertHardcopy', 'off')
```

By publishing your file with this window open and the **Use new figure** option set to false, any code-generated figure takes the properties of the open Figure window.

i Note

You must set the **Figure capture method** option to **entireFigureWindow** for the final published figure to display all the properties of the open Figure window.

Creating a Thumbnail

You can save the first code-generated graphic as a thumbnail image. You can use this thumbnail to represent your file on HTML pages. To create a thumbnail, follow these steps:

1. On the **Publish** tab, click **Publish** ▼ and select **Edit Publishing Options**. The Edit Configurations dialog box opens.
2. Set the **Image Format** option to a bitmap format, such as .png or .jpg. MATLAB creates thumbnail images in bitmap formats.
3. Set the **Create thumbnail** option to true.

MATLAB saves the thumbnail image in the folder specified by the **Output folder** option in the **Publish settings** pane.

Save a Publish Setting

You can save your publish settings, which allows you to reproduce output easily. It can be useful to save your commonly used publish settings.

Publish settings: User Default (modified) ▼ Save As...

Output settings	
Output file format	html
● Output folder	I:\my_MATLAB_files\my_published_files
XSL file	
Figure settings	
● Figure capture method	getframe
Image Format	default (png)
Use new figure	true
Max image width (pixels)	Inf
Max image height (pixels)	Inf
Create thumbnail	true
Code settings	
Include code	true
Evaluate code	true
● Catch error	false
Max # of output lines	Inf

When the **Publish settings** options are set, you can follow these steps to save the settings:

1. Click **Save As** when the options are set in the manner you want.

The **Save Publish Settings As** dialog box opens and displays the names of all the currently defined publish settings. By default the following publish settings install with MATLAB:

- Factory Default

You cannot overwrite the Factory Default and can restore them by selecting Factory Default from the **Publish settings** list.

- User Default

Initially, User Default settings are identical to the Factory Default settings. You can overwrite the User Default settings.

2. In the **Settings Name** field, enter a meaningful name for the settings. Then click **Save**.

You can now use the publish settings with other MATLAB files.

You also can overwrite the publishing properties saved under an existing name. Select the name from the **Publish settings** list, and then click **Overwrite**.

Manage a Publish Configuration

- [Running an Existing Publish Configuration](#)
- [Creating Multiple Publish Configurations for a File](#)
- [Reassociating and Renaming Publish Configurations](#)
- [Using Publish Configurations Across Different Systems](#)

Together, the code in the **MATLAB expression** pane and the settings in the **Publish settings** pane make a publish configuration that is associated with one file. These configurations provide a simple way to refer to publish settings for individual files.

To create a publish configuration, on the **Publish** tab, click **Publish ▼**, and select **Edit Publishing Options**. The Edit Configurations dialog box opens, containing the default publish settings. In the **Publish configuration name** field, type a name for the publish configuration, or accept the default name. The publish configuration saves automatically.

Running an Existing Publish Configuration

After saving a publish configuration, you can run it without opening the Edit Configurations dialog box:

1. Click **Publish** . If you position your mouse pointer on a publish configuration name, MATLAB displays a tooltip showing the MATLAB expression associated with the specific configuration.
2. Select a configuration name to use for the publish configuration. MATLAB publishes the file using the code and publish settings associated with the configuration.

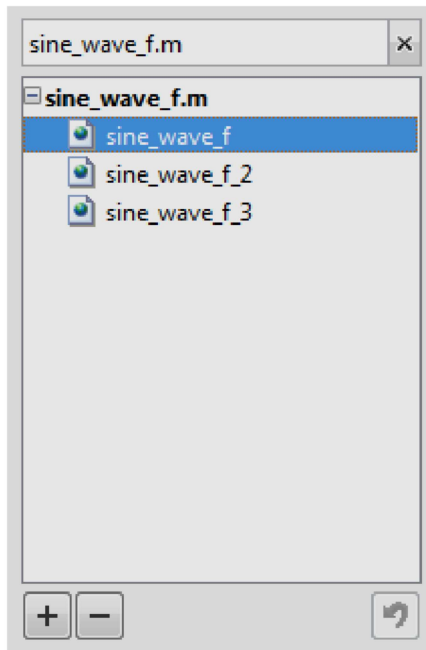
Creating Multiple Publish Configurations for a File

You can create multiple publish configurations for a given file. You might do this to publish the file with different values for input arguments, with different publish setting property values, or both. Create a named configuration for each purpose, all associated with the same file. Later you can run whichever particular publish configuration you want.

Use the following steps as a guide to create new publish configurations.

1. Open a file in your Editor.
2. Click the **Publish** button drop-down arrow, and select **Edit Publishing Options**. The Edit Configurations dialog box opens.
3. Click the **Add** button **+** located on the left pane.

A new name appears on the configurations list, *filename_n*, where the value of *n* depends on the existing configuration names.



4. If you modify settings in the **MATLAB expression** or **Publish setting** pane, MATLAB automatically saves the changes.

Reassociating and Renaming Publish Configurations

Each publish configuration is associated with a specific file. If you move or rename a file, redefine its association. If you delete a file, consider deleting the associated configurations, or associating them with a different file.

When MATLAB cannot associate a configuration with a file, the Edit Configurations dialog box displays the file name in red and a **File Not Found** message. To reassociate a configuration with another file, perform the following steps.

1. Click the **Clear** search button **x** on the left pane of the Edit Configurations dialog box.
2. Select the file for which you want to reassociate publish configurations.
3. In the right pane of the Edit Configurations dialog box, click **Choose....** In the Open dialog box, navigate to and select the file with which you want to reassociate the configurations.

You can rename the configurations at any time by selecting a configuration from the list in the left pane. In the right pane, edit the value for the **Publish configuration name**.

i Note

To run correctly after a file name change, you might need to change the code statements in the **MATLAB expression** pane. For example, change a function call to reflect the new file name for that function.

Using Publish Configurations Across Different Systems

Each time you create or save a publish configuration using the Edit Configurations dialog box, the Editor updates the `publish_configurations.m` file in your settings folder. (This is the folder that MATLAB returns when you run the `MATLAB prefdir` function.)

Although you can port this file from the settings folder on one system to another, only one `publish_configurations.m` file can exist on a system. Therefore, only move the file to another system if you have not created any publish configurations on the second system. In addition, because the `publish_configurations.m` file might contain references to file paths, be sure that the specified files and paths exist on the second system.

MathWorks® recommends that you not update `publish_configurations.m` in the MATLAB Editor or a text editor. Changes that you make using tools other than the Edit Configurations dialog box might be overwritten later.

See Also

Topics

[Publish and Share MATLAB Code](#)

[Publishing Markup](#)